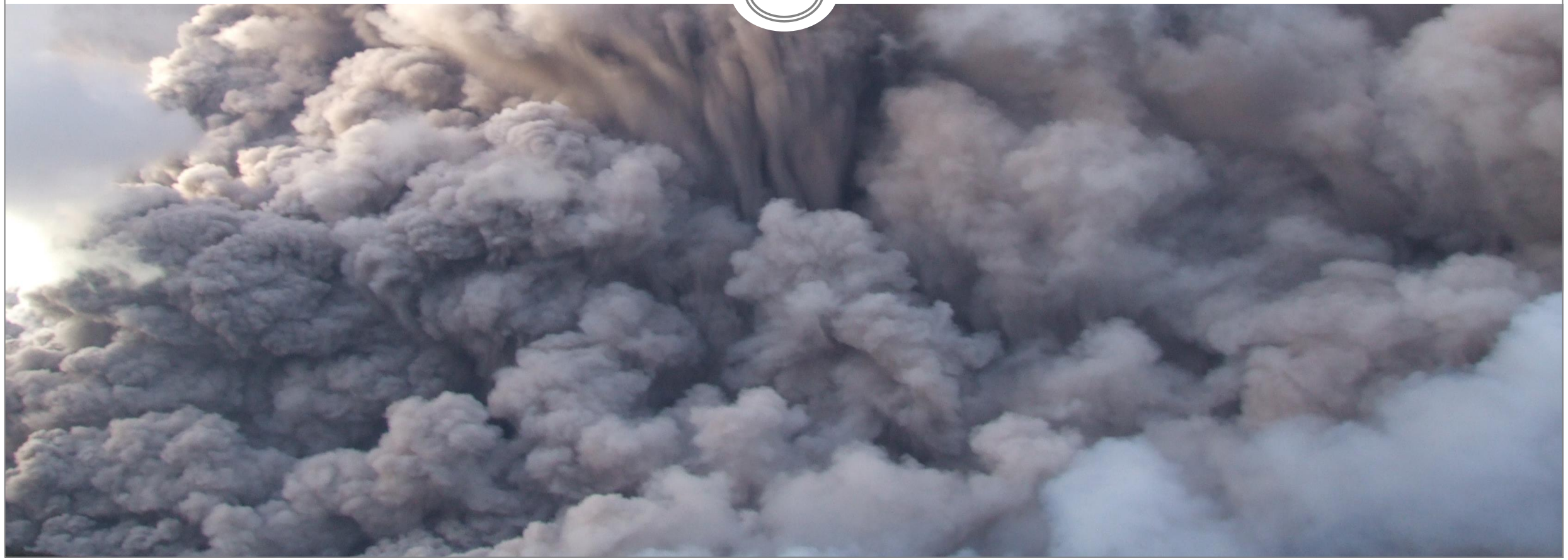
Choosing volumes: more details



LAHARZ equations are based on very well measured lahars, most of which are $10^4 – 10^9$ m^3 (10,000 m^3 – 1 km^3) in volume

(Iverson et al. 1998)

TABLE 1. BASIC DATA RANKED IN ORDER OF FLOW VOLUME

Name of event	Location and date	Data source	Flow volume, V (m^3)	Inundated cross section Area, A (m^2)	Inundated planimetric area, B (m^2)
Osceola	Mount Rainier, United States,. 5000 B.P	Vallance and Scott, 1997	4.0×10^9	1.5×10^5	5.5×10^8
Teteltzingo	Citlaltepetl, Mexico, 18000 B.P.	Carrasco-Núñez et al., 1994	1.8×10^9	N.D.	1.4×10^8
Electron	Mount Rainier, United States, 530–550 B.P	Crandell, 1971	2.5×10^8	3.7×10^4	6×10^7
Round Pass	Mount Rainier, United States, 2700 B.P	Crandell, 1971	2×10^8	N.D.	5×10^7
Dead Man Flat	Mount Rainier, United States, 1100 B.P.	K. Scott et al., 1995; J.W. Vallance, unpublished data	1.8×10^8	1.4×10^4	9.0×10^7
National	Mount Rainier, United States, 500–1800 B.P.	K. Scott et al., 1995; J.W. Vallance, unpublished data	1.5×10^8	1.0×10^4	7.8×10^7
Paradise	Mount Rainier, United States, 4500–5000 B.P.	Crandell, 1971	1×10^8	N.D.	3.4×10^7
Zigzag	Mount Hood, United States, 1700 B.P	J. W. Vallance, unpublished data	7.3×10^7	1.2×10^4	5.5×10^7
Trout Lake	Mount Adams, United States, 6000 B.P.	Vallance, 1998	6.6×10^7	N.D.	2.7×10^7
Middle Fork Nooksack	Mount Baker, United States, 6000 B.P.	Hyde and Crandell, 1978	5×10^7	N.D.	2×10^7
Kautz Creek	Mount Rainier, United States, 1947	Crandell, 1971; Scott and Vallance, 1995	4×10^7	5000	4.5×10^6
Azufrado	Nevado del Ruiz, Columbia, 1985	Fritz et al., 1986; Pierson et al., 1990	4×10^7	2300	3.4×10^7
Molinos Nereidas (Chinchina)	Nevado del Ruiz, Columbia, 1985	Fritz et al., 1986; Pierson et al., 1990	3×10^7	1100	6.0×10^6
Guali	Nevado del Ruiz, Columbia, 1985	Fritz et al., 1986; Pierson et al., 1990	1.6×10^7	2000	1.1×10^7
Salt Creek	Mount Adams, United States, 200 B.P.	Vallance, 1998	1.5×10^7	N.D.	1.6×10^7
Tahoma	Mount Rainier, United States, 400–500 B.P.	Scott et al., 1995	1.5×10^7	1.9×10^4	6.0×10^6
Pine Creek + Muddy River	Mount St. Helens, United States, 1980	Pierson, 1985	1.4×10^7	2100	1.8×10^7
South Fork Toutle	Mount St. Helens, United States, 1980	Janda et al., 1981; Fairchild and Wigmosta, 1983	1.2×10^7	1500	3×10^7
Whitney Creek	Mount Shasta, United States, 1935	Osterkamp et al., 1986	4×10^6	N.D.	8×10^6
Bolum Creek	Mount Shasta, United States, 1897	Osterkamp et al., 1986	1.5×10^6	N.D.	3×10^6
Mabinit Eruption Lahars	Mayon, Philippines, 1984	Rodolfo, 1989	1.2×10^6	200	1.8×10^6
Tahoma Creek	Mount Rainier, United States, 1988	Walder and Driedger, 1994; J.S. Walder, unpublished data	6×10^5	190	1×10^6
Blue Lake	Mount St. Helens, United States, 1980	Major, 1984; Major and Voight, 1986	3.8×10^5	320	7.5×10^5
Butte Canyon	Mount St. Helens, United States, 1980	Major, 1984; Major and Voight, 1986	3.8×10^5	300	5.0×10^5
Mabinit Typhoon Saling	Mayon, Philippines, 1985	Rodolfo, 1989; Rodolfo et al.,1989	3×10^5	200	2×10^5
Middle	Mount St. Helens, United States, 1980	Major, 1984; Major and Voight, 1986	1.4×10^5	N.D.	4.0×10^5
Polallie Creek	Mount Hood, United States, 1980	Gallino and Pierson, 1984; T.C. Pierson, unpublished data	8×10^4	300	4.7×10^5
West Dodson	Columbia Gorge, Oregon, United States, 1996	R. M. Iverson, unpublished data	8×10^4	90	1×10^5
Mayflower Gulch	Tenmile Range, Colorado, United States, 1961	Curry, 1966	1.7×10^4	30	1.6×10^4

How big, how far, how frequent?

- Small: Up to 10^4 m^3, up to 10 km, once or more per decade
- Large: 10^4 to 10^6 m^3, up to 50 km, once or more per century
- Very large: 10^6 to 10^8 m^3, up to 100 km, once per 100 to 1,000 years
- Extremely large: 10^8 to 10^{10} m^3, >200 km, once per 1,000 to 10,000 years

10^5 - 10^6 m^3





Semeru, Indoneisa, February 2007,
lahars range **<u>100,000 – 1,000,000 m^3</u>**





Atacama, Chile, March 2015
Estimated <u>2 million m^3</u>

10^5 - 10^6 m^3





Ruapehu in NZ March 2007

~<u>7-9 million m^3</u>

Mayon, Philippines, 2006 (typhoon Reming), lahar volumes range **<u>300,000 m^3 to 9 million m^3</u>**

Table 2 Calculated lahar volumes and deposit thicknesses exposed in gullies on the eastern and southern flanks of Mayon volcano

Volume calculation and deposit thickness

Area	Lahar deposit type	Max mapped thickness (m)	Min mapped thickness (m)	Volume (m^3)
Guinobatan–Camalig	Debris flow to hyperconcentrated flow	5.0	0.3	8,700,000
Daraga–Legaspi	Debris flow to hyperconcentrated flow	5.0	0.3	7,850,000
Padang	Debris flow to hyperconcentrated flow	1.6	0.1	850,000
Lidong	Debris flow to hyperconcentrated flow	1.5	0.2	300,000
San Antonio	Debris flow to hyperconcentrated flow	1.7	0.2	1,100,000
Total				18,800,000

10^7 m^3



Mount Merapi, Indonesia, October 2010, lahars in 8 channels flowed 15-20 km & range **8-28 million m³**

Lahar floods continue to threaten people who live along the rivers upstream from the slopes of Mount Merapi. The eruption material of Mount Merapi was predicted to attain 130 cubic meters and is concentrated in eight of the rivers upstream as shown in table 1.

Table 1: The eruption material of Mount Merapi which was concentrated in 8 surrounding Lahar Rivers (Mujiharjo *et al*. [1]).

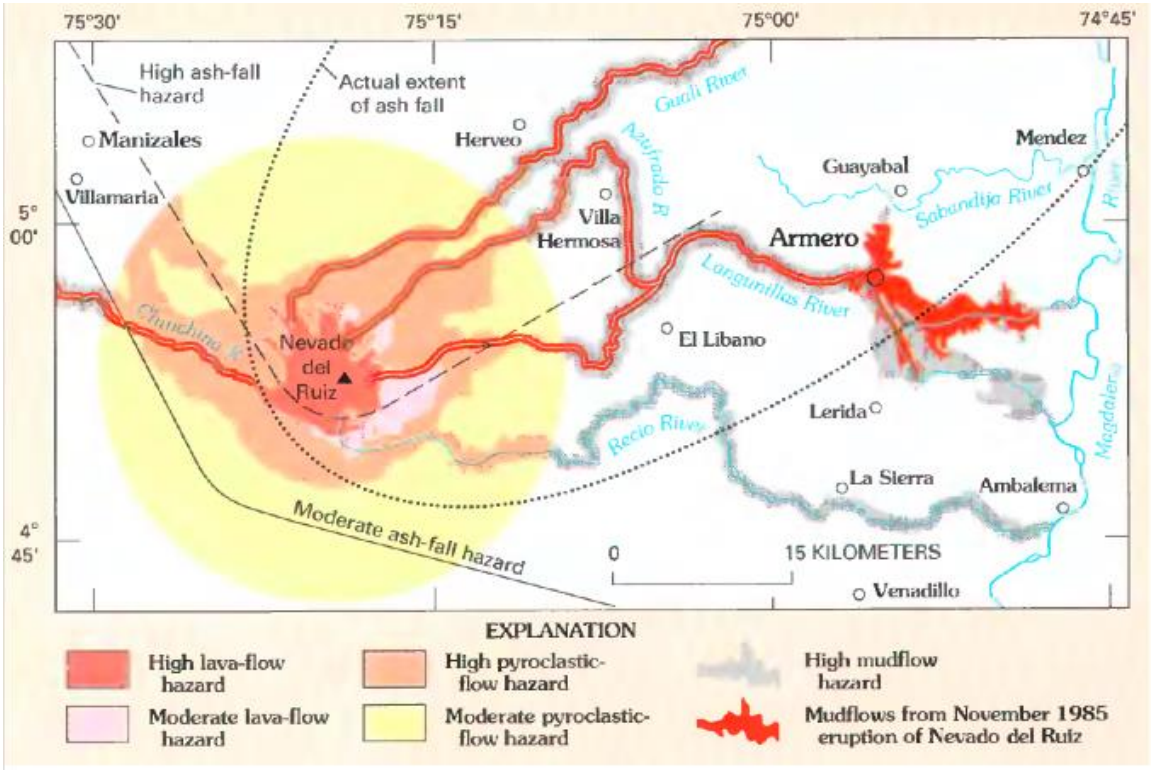
No.	Lahar River	Pyroclastic deposit amount (m^3)
1	Boyong	8 million
2	Bebeng	10 million
3	Blongkeng	10 million
4	Woro	12 million
5	Kuning	14 million
6	Putih	18 million
7	Pabelan	24 million
8	Gendol	28 million
Total		124 million

10^7 m^3



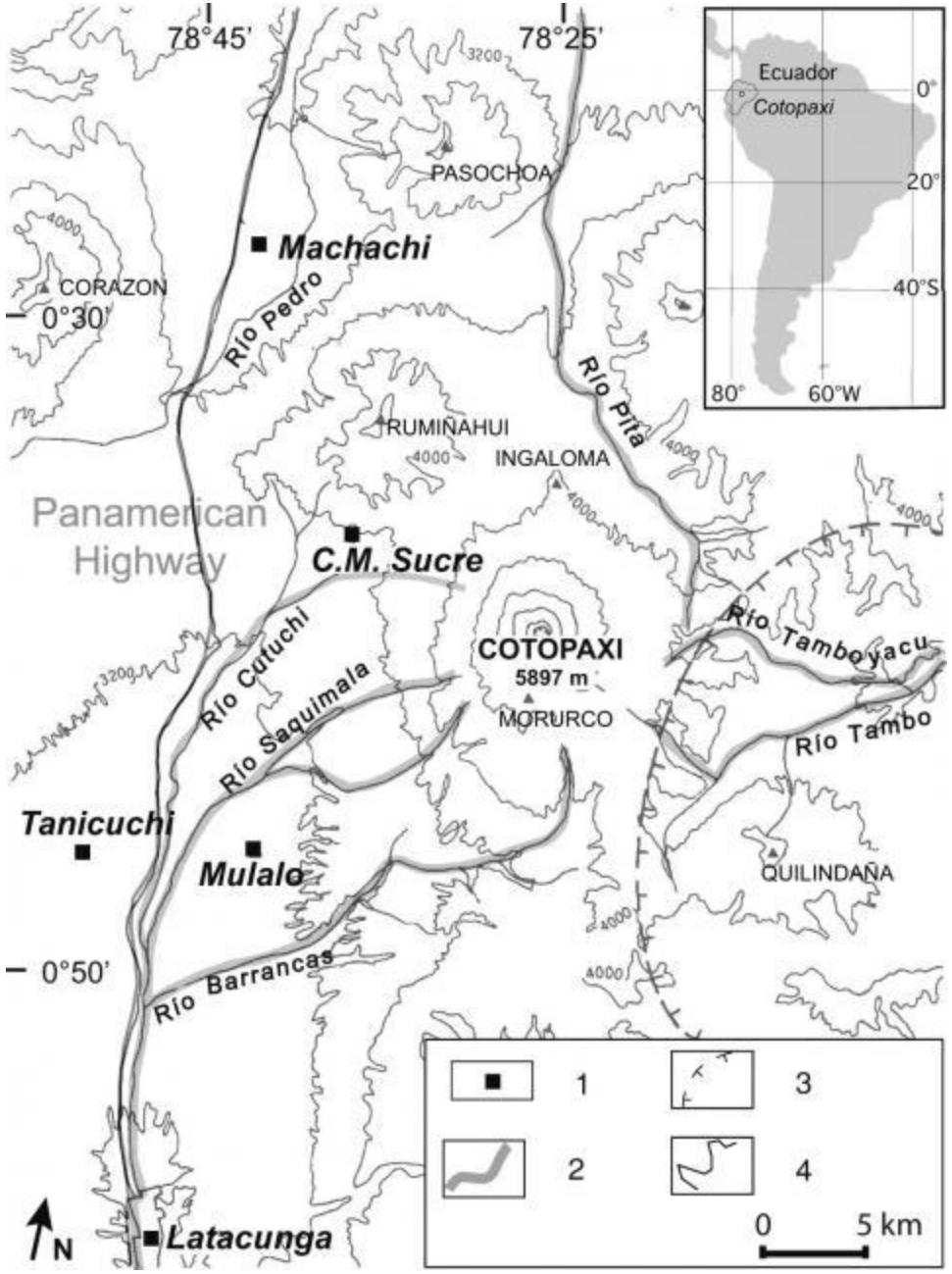


Armero Tragedy
>25,000 deaths of 30,000 inhabitants



Nevado del Ruiz, Colombia, 1985
Lahars in 3 channels range from
40 million m^3 (Azufrado)
30 million m^3 (Molinos Nereidas- Chinchina)
16 million m^3 (Guali)

10^8 m^3

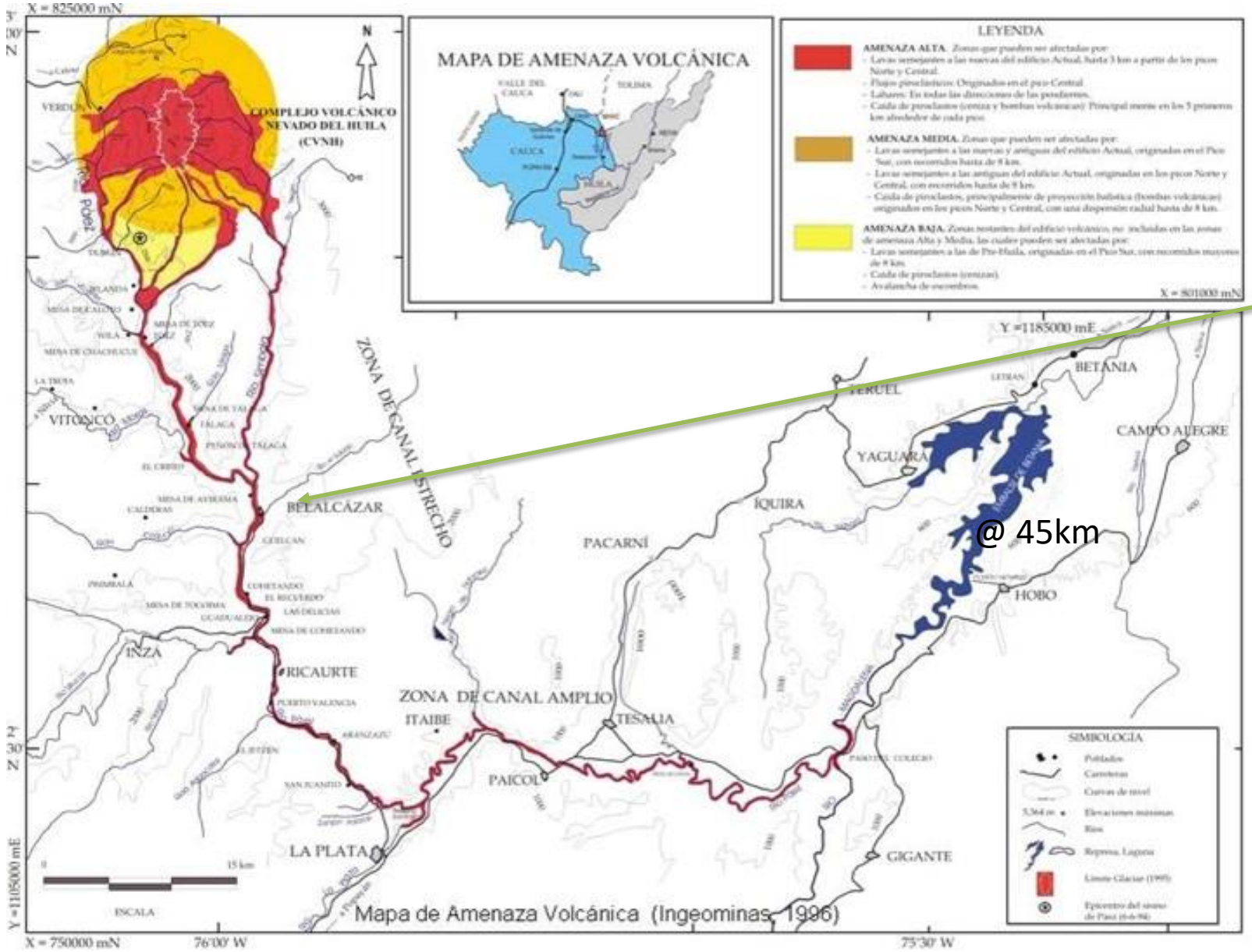


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Cotopaxi, Ecuador, 1877 lahars
Traveled 100 km
Total volume of **0.5 km^3**
Accumulated from 4 drainages

10^8 m^3





Nevado del Huila, Colombia, February 2007, lahars traveling ~150 km with a volume **<u>0.3 km³</u>**





10^9 m^3



Mount Rainier Osceola Lahar, USA, 5600 years ago, 3 lahars traveling ~100 km covering 550 km^2 with a combined volume of **4 km^3**

How to choose volumes for LAHARZ

- Because of uncertainty and naturally occurring error in the regression equations, you should choose volumes that increase by at least 2x (1, 2, 4, 8, 16 million m^3, for example)
- Even better is to use values that increase by $10^{0.5}$ (1, 3, 10, 30, 100 million m^3, for example)

How to choose volumes for LAHARZ

- Check historical records of lahar volumes at your volcano
  - What types of lahars are produced at your volcano?
    - Rainfall-triggered
    - eruption-triggered (Snow melt? Ice melt? Groundwater expulsion? Damming of rivers by deposits? Flows impacting lakes or rivers? Crater lake expulsion?)
  - What is the most common lahar volume?
  - What is the worst-case scenario (biggest) lahar volume?
- Investigate the amount of snow and ice on your volcano
- Investigate whether any crater lakes are present
- Investigate the rainfall/typhoon/hurricane history at your volcano
  - Is there a rainy season?
- Check historical records at analog volcanoes
  - Volcanoes of a similar size
  - Volcanoes with similar amounts of snow/ice
  - Volcanoes with similar eruptive histories
  - Volcanoes with similar crater lakes
  - Volcanoes in similar climates